Choose a motion method

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose a motion relationship and justify its assumptions.

SUCCESS CRITERIA - I can...

1. Choose a relationship using the known quantities.
2. Show linked calculations with units and directions.
3. State when constant velocity or acceleration applies.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “constant” and “interval” to explain your model or result. Sentence starters are optional.

constant: not changing during the stated interval

interval: the time or range between two stated values

acceleration: rate of change of velocity

01 Warm-up: bring back an earlier idea

x_i=-3±0.5 m, v=5±0.1 m/s and exact t=7 s.
Give central final position, its conservative interval and full width.
Name an assumption.

Word help: final = at the end of the interval; subscript f

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

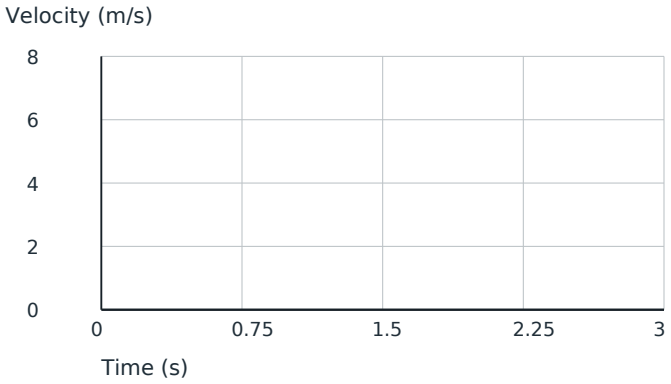
02 Worked example: explain the model

Reference: $v_f = v_i + a\,\Delta t$; $\Delta x = v_i\,\Delta t + \frac{1}{2}a(\Delta t)^2$
$v_f^2 = v_i^2 + 2\,a\,\Delta x$.
Constant acceleration; i = initial, f = final. Choose the equation containing your knowns and unknown.

Worked reasoning: From rest, a cart accelerates uniformly at 2 m/s² for 3 s. Final velocity=6 m/s; average velocity=(0+6)/2=3 m/s; displacement=3×3=9 m.

Your turn: Complete the cart's velocity-time graph. Shade the triangular area under the line and use it to check the 9 m displacement.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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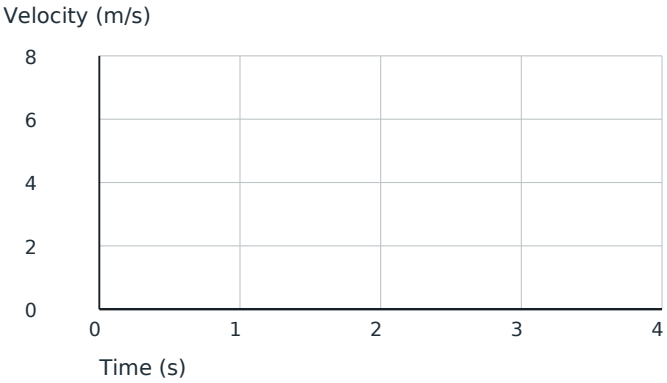
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Solve and check

A cart starts at the -3 m mark on a test track, moving at +2 m/s. It accelerates at a constant +1 m/s² for 4 s.
Draw its velocity-time graph, then find final velocity, displacement and final position.

Sentence starter: I found ... [unit]. This value means ... in this situation.

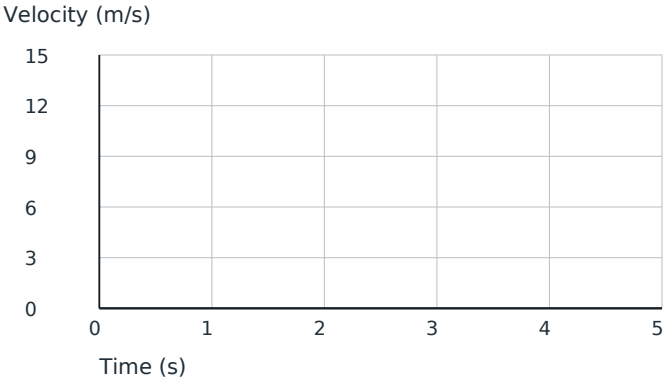


Marzano check - circle: 1 2 3

04 No time given

A cyclist brakes from 12 m/s to rest with constant acceleration -3 m/s². Forward is positive.
Find stopping distance without first calculating time. Then find the stopping time and check with a velocity-time graph.

Sentence starter: I found ... [unit]. This value means ... in this situation.

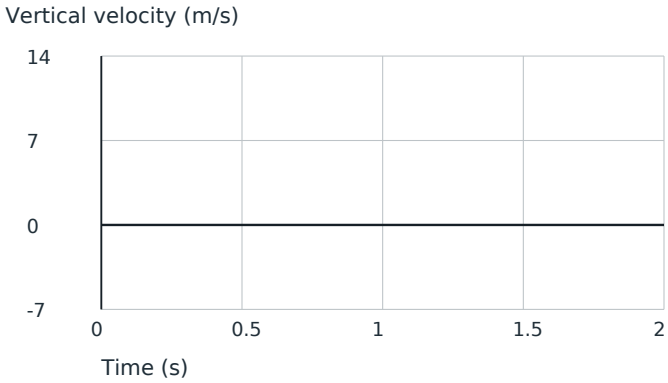


Marzano check - circle: 1 2 3

05 At the top

You toss a ball straight upward at 14 m/s. Ignore air resistance and use g = 9.8 m/s².
Draw velocity and acceleration arrows at release and at the highest point.
Find the maximum rise and time to the top. Is acceleration zero there?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

Choose a motion method

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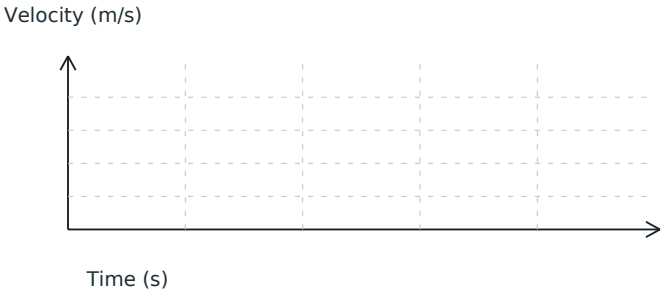
Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Velocity changes from -3 to -15 m/s in 3 s, then stays at -15 m/s for 7 s.
Find acceleration and describe the speed trend in each interval.
Sketch the velocity-time model; label the given values.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A car starts at +12 m/s and brakes with a=-3 m/s² until it stops. A student extends the equation to 8 s and reports v=-12 m/s.
Explain what assumption fails and give stopping time/distance.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

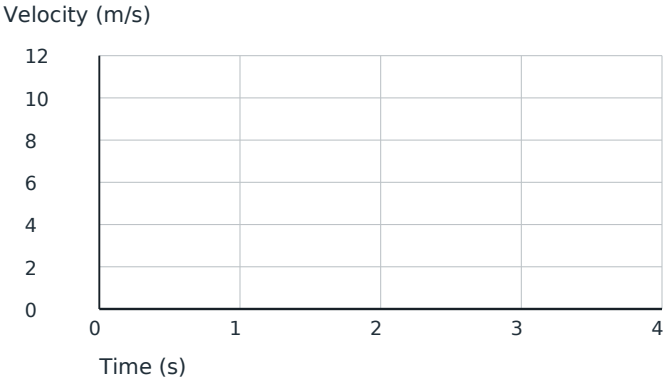
Marzano check - circle: 1 2 3

08 STRETCH: test the model

A cart starts at 2 m/s and travels 24 m in 4 s with uniform acceleration. Solve for acceleration in two ways and explain why a constant-speed method fails.

Word help: uniform = constant throughout the stated interval

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: 1 2 3